REPORT 5 - NOSE CONE 720 MM LOADS FOR STRUCTURE

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Date: November 1, 2026

Doc ID: 2027_C_FD_REPORT-5_NOSE-CONE-720MM

Weisshorn – 720 mm Nose Cone: Pressure, Forces and Heating

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This report gives ST the aerodynamic and thermal inputs to size the two 720 mm noses. ST computes the joint and shell loads from them. Both noses are 720 mm long with $D = 180$ mm and a 4.5 mm tip radius; their geometry is in `2027_C_FD_NOSE-CONE-720MM_GEOMETRY`.

- **A** (baseline, 3U inside the nose): LV-Haack, 679.0 mm + 41 mm coupler.
- **B** (backup, 3U elsewhere): Von Kármán, 700.0 mm + 20 mm coupler.

The numbers come from eight Fluent runs (2D axisymmetric RANS, zero incidence) at $M = 0.95$, 1.20 and 1.47 (burnout), and from the Python flight model, calibrated on these runs, for the time histories and the hot day (30 °C, LVO masses). The nose is **sealed**: the air inside stays at the pad pressure $p_{pad} = 101.3$ kPa, raised by its heating,

$$p_{in} = p_{pad}\,\frac{T_{in}}{T_{pad}}, \qquad \Delta p = p_w - p_{in}, \tag{1}$$

between $T_{in} = T_{pad}$ and T_{in} = mean temperature of the inner face of the shell (upper bound). $\Delta p > 0$ pushes the shell inwards. Axial forces are positive when they push the nose aft onto the joint (compression).

Table 1: Load cases. Pressures: tip with p_{in} at T_{pad} (largest inward load), shell minimum with heated air (largest outward load). Net pressure force on the joint: $\sum (p_w - p_{in})\, 2\pi r\, \mathrm{d}r$ over the outer surface, for $T_{in} = T_{pad}$ / heated air. D_n: aerodynamic axial force of the nose (model, 20 µm roughness). N/α: normal force per degree of incidence.

	A nominal	A hot day	B nominal	B hot day
LC1, burnout (M_{max}, q_{max})				
Time t [s], Mach M	9.1, 1.47	9.1, 1.56	9.1, 1.47	9.1, 1.56
Dynamic pressure q [kPa]	116.8	131.4	117.5	132.4
Δp at the tip [kPa]	153	175	154	177
Lowest Δp on the shell [kPa]	-46	-49	-48	-50
Net pressure force on the joint [N]	-397 / -753	-374 / -767	-436 / -793	-418 / -813
Aerodynamic axial force D_n [N]	278	310	234	260
Normal force N/α [N/°]	104	117	104	118
LC2, transonic ($M = 0.95$ on the ascent)				
Time t [s], q [kPa]	5.9, 57.6	5.8, 57.9	5.9, 57.3	5.8, 58.0
Δp at the tip [kPa]	63	64	62	64
Lowest Δp on the shell [kPa]	-22	-21	-29	-28
Net pressure force on the joint [N]	-265 / -358	-248 / -335	-263 / -355	-246 / -333
Aerodynamic axial force D_n [N]	53	51	44	47
Normal force N/α [N/°]	51	51	51	52
LC3, apogee ($q \approx 0$)				
Time t [s], p_∞ [kPa]	43.7, 31.3	44.2, 31.1	43.9, 30.8	44.5, 30.5
Δp on the whole shell [kPa]	-70	-70	-71	-71
Net pressure force on the joint [N]	-1783	-1788	-1794	-1802
Nose centre of pressure x_{CP} from the tip [mm]	295		347	

Table 2: Heating. Fluent values: burnout, nominal, wall at 300 K. Maximum temperatures over the flight: Python model calibrated on Fluent (Section **??**), CFRP shell 1.2 mm, Al 6082 tip insert 80 mm.

	A nominal	A hot day	B nominal	B hot day	Limit
Heat flux at the tip q_w [kW/m^2] (Fluent)	172.5	–	173.4	–	
h at the tip / shoulder [W/(m^{2}K)] (Fluent)	1888 / 546	–	1899 / 539	–	
Stagnation temperature T_0 [°C]	118	154	119	156	
Al tip insert [°C]	61	84	56	79	
Shell outer face, 50 mm [°C]	82	110	82	110	120 (DP490)
Shell inner face, 50 mm [°C]	77	104	78	104	67 (Req_10)
Inside air, upper bound [°C]	72	98	73	99	67 (Req_10)

What this means for the structure.

1. **The sealed nose is the largest load.** At apogee the outside pressure has dropped to 31.3 kPa while the inside stays at the pad pressure: $\Delta p \approx -70$ kPa on the whole shell, which pulls the nose forward off the joint with about 1.8 kN, close to the 2000 N of Req_07. At burnout the inside overpressure already outweighs the drag: the net pressure force is -397 to -753 N (tension). Vent holes would bring p_{in} to p_∞ and remove this load.

2. **Inward pressure** is highest at the tip at burnout, 153 kPa (175 kPa hot day), on the spherical cap only. Beyond $x = 2$ mm it is at most 36 kPa for A and 17 kPa for B (43 and 22 kPa hot day), and below 10 kPa after 50 mm (Figure **??**).

3. **Aerodynamic axial force** peaks at burnout: 278 N for A and 234 N for B (310 and 260 N hot day). Fluent with a smooth wall gives 282.6 and 237.6 N.

4. **Normal force** under incidence α: $N = 2\,q\,A_{ref}\,\alpha$, acting at x_{CP}, distributed as $\mathrm{d}N/\mathrm{d}x = 2\,q\,\alpha\,\mathrm{d}S/\mathrm{d}x$ (Section **??**). ST chooses the design incidence (Req_12).

5. **Temperatures.** The DP490 limit (120 °C) is not reached. The inner face of the shell and the inside air exceed Req_10 (67 °C) already on a nominal day, for both designs.

6. **A against B.** The heating is the same within 1 %. B has 15 % less drag at burnout, about half the inward pressure on the front of the shell (item 2), and a 20 mm coupler instead of 41 mm. The sealed-nose loads are the same for both.

   The data behind every figure are delivered as CSV tables (Appendix **??**).